

Venture Capital and Entrepreneurial Management

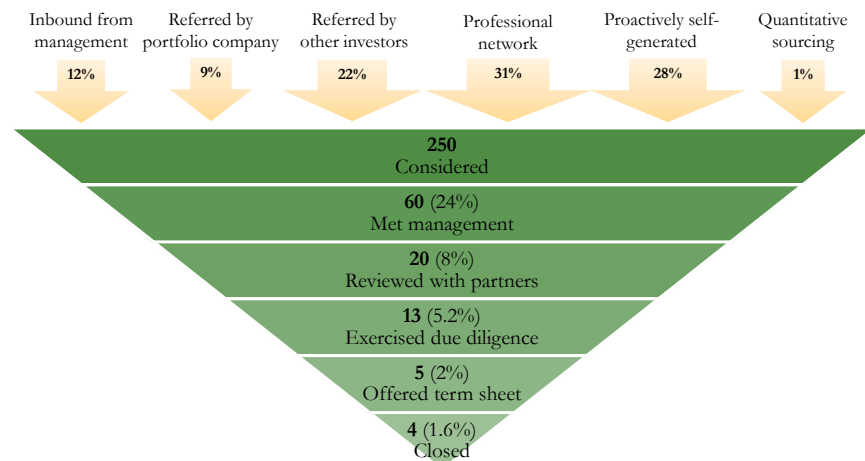
MGMT 264

Opportunity Evaluation

Raphael ("Raffi") Amit, PhD
The Marie and Joseph Melone Professor
The Wharton School
University of Pennsylvania
☎ (215) 898 7731
✉ amit@wharton.upenn.edu

Evaluating Opportunities

- The VC Funnel of Early Stage VCs



Source: Gompers et al, Working Paper (2016)

Evaluating Opportunities

- Which idea looks like a good **business opportunity** for you?



A social app which enables users to send their friends the word "yo" as a text and audio notification (2014)



Peer-to-peer marketplace for storage and parking that transforms your empty space (e.g. basements, offices) into rentable self-storage and parking space (2007)



Online business that sells pet accessories and supplies to retail customers (1998)



\$15 computer with a 1.2Ghz board, 512MB SDRAM, 10/100Mbps Ethernet Port, 3.5 MM Audio/Mic Output, 4K HDMI and 2 USB ports (2016)

An idea that is feasible, desirable, scalable, viable (i.e., profitable)

Entrepreneurial Opportunities

- Management vs. Market

- The "Jockey vs. Horse" Debate

- A good opportunity/business (i.e. good market and technology)
- A good entrepreneur/entrepreneurial team



- A very classic debate among VCs: **Which one is more important? Or which one will you bet on?**



VS.



Entrepreneurial Opportunities

- Market vs. Management

■ “Betting on the Jockey”

□ Research:

- 4 out of the top 5 criteria most frequently rated essential are related to entrepreneurs!
- Around one-half of venture capitalists will not even consider a venture that does not have a balanced team.

□ Practice:

- George Doriot, “father of Venture capitalism”:
 - “Always consider investing in a grade A man with a grade B idea. Never invest in a grade B man with a grade A idea.”

Criteria Most Frequently Rated Essential	Percentage
Capable of sustained intense effort	64%
Thoroughly familiar with market	62%
At least ten times return in 5-10 years	50%
Demonstrated leadership in past	50%
Evaluates and reacts to risk well	48%
Investment can be made liquid	44%
Significant market growth	43%
Track record relevant to venture	37%
Articulate venture well	31%
Proprietary protection	29%

Source: MacMillan et al, Journal of Business Venturing (2005)



Entrepreneurial Opportunities

- Market vs. Management

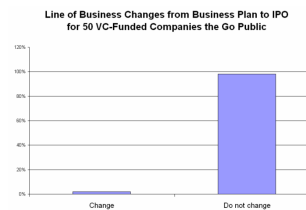
■ “Betting on the Horse”

□ Research:

- More than 90% of companies did not change their line of business or business focus
- Founders were only CEOs in 58% of companies
- **Initial business lines are core attributes.**

□ Practice:

- Don Valentine founder of Sequoia Capital, investor in Cisco, assessed the market for the product or service.
 - Cisco was turned down by many other VCs because the team was considered weak.
 - Valentine invested in Cisco anyway. He saw a huge market.



Source: Kaplan et al, NBER (2005)



Entrepreneurial Opportunities

- Market vs. Management

■ The “Jockey vs. Horse” Debate

Investment Theses and Risks in Venture Capital Analyses	Reason to Invest/Strength	Risk of Investment/Weakness
Quality of management	59.7%	61.2%
Performance to date	26.9%	7.5%
Funds at risk/downside	19.4%	13.4%
Influence of other investors	6.0%	6.0%
VC portfolio fit and monitoring cost	17.9%	14.9%
Valuation	20.9%	19.4%
Market size and growth	68.7%	31.3%
Competition and barriers to entry	32.8%	40.3%
Likelihood of customer adoption	29.9%	22.4%
Financial market and exit conditions	16.4%	7.5%
Product and/or technology	40.3%	31.3%
Business strategy/model	53.7%	50.7%

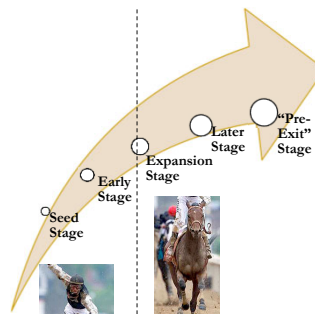
Source: Kaplan & Stromberg, Journal of Finance (2004)

Entrepreneurial Opportunities

- Market vs. Management

■ The “Jockey vs. Horse” Debate

- Different philosophies with equifinality!!
(a concept borrowed from General System Theory which in the context of a venture suggests that similar results may be achieved with different initial conditions and in many different ways)
- Might depend on the stage of the venture as well as the investment time horizon.
- Might lead to distinct deal structures as well as post-investment management strategies.



Evaluating Opportunities

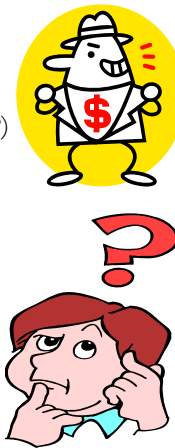
■ A Simplified View – MMM Framework



Entrepreneurial Opportunities

Management Related Factors– MMM Framework Investors' Lens

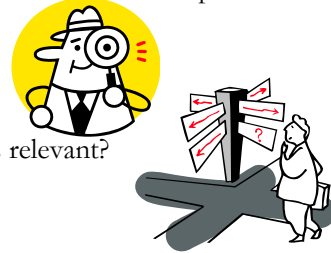
- Management Team
 - ❑ Can management make it happen? Vision? Execute? Grow the company?
 - ❑ Experience? Reputation? Is it complete? (breadth, depth?) Track record? Do they complement each other (in skills and personality)? Relationship among the founders? Passion?
 - ❑ Do we trust them? Can we work with them?
 - ❑ Execution risks?
 - ❑ Do they have good chemistry, trust and mutual respect among them?
 - ❑ What do the equity splits and governance structures look like?



Entrepreneurial Opportunities

Management Related Factors– MMM Framework Entrepreneurs' Lens

- ❑ What do we bring to the table that matters in this opportunity?
- ❑ Why us? Can we execute? Are we getting along? Trust? Respect? Shared values?
- ❑ What are our defined roles and responsibilities moving forward?
- ❑ Is there a fit between our skills/experiences/interests and the perceived market opportunity?
- ❑ What skills are we missing?
- ❑ Do we have the technical competencies?
- ❑ Are our industry experiences & relationships relevant?
- ❑ Does our track record measure up?
- ❑ Are we truly committed? Passionate?



Entrepreneurial Opportunities

- Management Related Factors

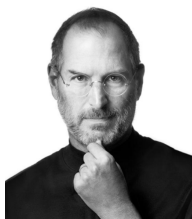
- Entrepreneur – Which profile would you invest?

A



- Started companies in their 20s
- In good financial status
- Dropped out college earlier (smart though)
- No working experience

B



- Started companies in their 20s
- In poor financial status
- Learning by doing
- Very little working experience

C



- Started companies in their 30s or 40s
- In good financial status
- Well educated in college
- Several years working experience

Entrepreneurial Opportunities

- Management Related Factors

- Large Sample Research shows:
 - Around 90% of entrepreneurs were employed before^{1,2}.
 - Around 80% of entrepreneurs hold an advanced degree (e.g. graduate)¹.

Company	Number of Spawned Entrepreneurial Teams	Company	Number of Spawned Entrepreneurial Teams
IBM	70	Nortel Networks	17
AT&T	60	Unisys	17
Sun Microsystems	55	National Semiconductor	17
Apple Computer	48	Advanced Micro Devices	16
Hewlett-Packard	46	Mentor Graphics	15
Oracle	41	Cirrus Logic	14
General Electric	37	Genentech	14
Microsoft	30	Bristol Myers Squibb	14
Xerox	29	Raychem	14
Baxter	29	Dun & Bradstreet	14
Intel	28	SmithKline Beecham	13
Disney	28	MCI	13
Silicon Graphics	27	Pfizer	13
Lotus Development Corporation	26	Texas Instruments	13
Motorola	25	Lagard	12
DEC	24	Tandem Computer	12
Cadence Design Systems	21	Bank of America	11
Johnson & Johnson	21	PeopleSoft	11
Verizon	19	Morgan Stanley	11
Novell	18	Adaptec	11

Findings¹:

- Entrepreneurs with advanced degrees establish firms with innovation strategies.
- Entrepreneurs with senior management experience are more likely to obtain external financing (more legitimacy)

Source:

1. Burton et al, Research in the Sociology of Organizations (2002)
2. Gompers et al, Journal of Finance (2005)

Entrepreneurial Opportunities

- Management Related Factors

■ Entrepreneur



Entrepreneurial Opportunities

Market related factors - MMM Framework:

Investors' Lens

■ Product Markets

- ❑ Proven, large, accessible and attractive market? Size? Growth potential? Cyclical? Seasonal? Attractive industry?
- ❑ Innovative business model? Scalable? Robust? Flexible? Revenue model?
- ❑ Business Strategy? Why will they win? Competitive advantage?
- ❑ What are the Market risks? Technology risks?
- ❑ Product/service characteristics fit the perceived opportunity? Feasibility?
- ❑ Solves a real problem; strong motivation to buy... Value Proposition?
- ❑ Product can be differentiated?
 - Difficult to copy?
 - Is it a foundation of a platform for launching other products and services?
 - Can be designed and produced to realize high gross margins?
 - Is this a product or a business idea? Is the idea to develop a big company?
 - What distribution channels will be used?



Entrepreneurial Opportunities

Market related factors -- MMM Framework

Entrepreneurs' Lens

■ Product Markets

- ❑ What problem are we solving? What need are we satisfying? For whom?
- ❑ What is our value proposition to customers? Investors? Partners? Employees?
- ❑ What products/services should we offer?
- ❑ Market Size? Growth? Seasonal? Cyclical?
- ❑ How do we do business? (our business model)?
- ❑ How will we capture the market opportunity? (product/market strategy? Mode of entry? Sustainability? Scale? Scope?)
- ❑ Can we access the market? How? What channels should we use to serve customers in the most profitable manner?
- ❑ Competition? With whom? How?
- ❑ Why now? Are we too early? Too late? How fast?
- ❑ How will this opportunity change over time?
- ❑ Market structure: Barriers to entry? Patents? Barriers to Imitation? Buyer & supplier power? Concentration? Substitutes? Complementors? Exit barriers?



Evaluating Opportunities

- MMM Framework: Market-related factors

■ Market Size



□ Market Size Fallacy: An Example

- “Lending Club cites the size of the overall consumer credit market at **\$3.2 trillion** in the filing and said that **roughly \$380 billion of that would qualify under its relatively stringent qualifications for borrowers.**”
- “For the first six months of 2014, Lending Club originated \$1.8 billion in loans, generating \$86.9 million in revenue on that, up from \$718 million of loans originated and \$37.1 million in revenue for the same period of 2013.”

- Market size is defined by your value chain position (or business model) and customers’ willingness to pay...

Evaluating Opportunities

- MMM Framework: Market-related factors

■ Competition

“You must study the endgame before everything else”



-- José Raúl Capablanca
World Chess Champion

□ Critical Questions to ask:

- What if many other start-ups enter the market segment which might eliminate your projected profits?
- What if established players who are more resourceful decide to enter the market segment?
- What are the market (or business model) characteristics on which sustained competitive advantage could be built?
- How much does speed and scale matter in this market?

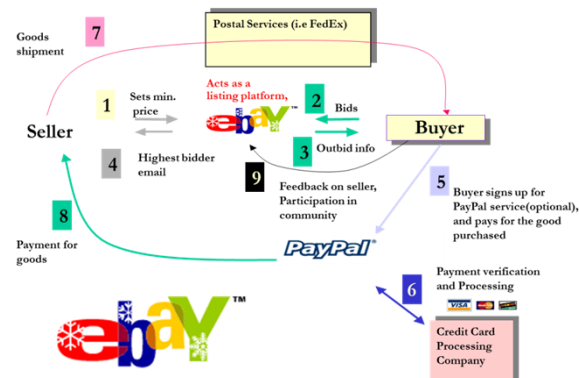
Entrepreneurial Opportunities

- MMM Framework: Market related factors

■ Market – Business Model

- E.g. How do you design your activity system to create value for all stakeholders?

Example: Business Model of ebay



Source: Amit and Zott, Strategic Management Journal (2001)

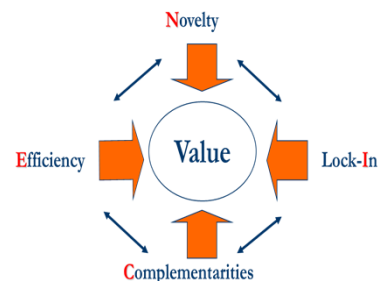
Evaluating Opportunities

- MMM Framework: Market-related factors

■ Business Model

- Critical Questions to ask:
 - What activities need to be enabled by the business model in order to create and deliver the offerings to the customers?
 - How are the activities linked? (sequencing, structure?)
 - Who are the stakeholders? Who is performing each activity in the business model?
 - Why is the business model creating value for all stakeholders and how much value can be appropriated (revenue model, cost model)?
 - Why is this business model better than the traditional ones?
 - Are there bottlenecks for scaling up in your business model?
 - What are the resources and capabilities that enable the business model?

NICE Value Drivers of Business Model



Source: Amit and Zott, Strategic Management Journal (2001)

Evaluating Opportunities

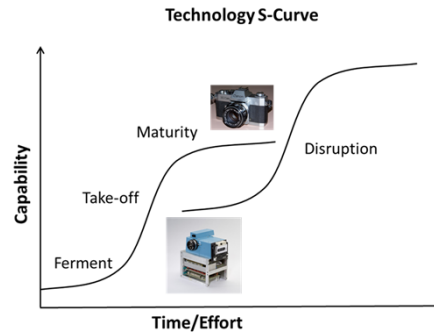
- MMM Framework: Market-related factors

■ Exploiting the Trends

- Economic
- Social/Demographic
- Technological

■ Timing

- Timing on the Demand Side
 - Are customers ready?
- Timing on the Supply Side
 - Is the eco-system ready?
- Timing on the Competition Side
 - Is this a good time to disrupt?



Entrepreneurial Opportunities

Money related Factors -- MMM Framework Investors' Lens



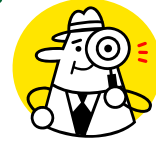
■ Money

- Investment
 - Financial characteristics- cost structure, investment required to reach CF breakeven?
 - How much money is needed for the next major milestone? To liquidity?
 - Can investment be staged?
 - Expected return (How big is the upside? Is it large enough to justify the financial risk? Does it fit my portfolio?)
 - Can I add value? How?
- Deal
 - Is the Valuation reasonable?
 - What are the terms? Downside protection (protective provisions?)
 - What are the main risks? Could I manage the risks with terms?
- Exit
 - IPO prospect
 - M&A
 - Time to liquidity

A Good Business Opportunity
is **NOT** necessarily
A Good Investment Opportunity

Entrepreneurial Opportunities

Money related Factors -- MMM Framework Entrepreneurs' Lens



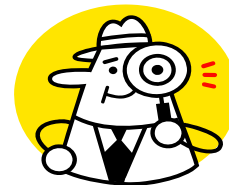
■ Money

- ❑ Investment needed to get to first milestone?
- ❑ Investment needed to get to cash flow break even?
- ❑ What is our revenue model? How will we make money?
- ❑ Will the expected return exceed our combined opportunity cost?
- ❑ Payback to investors?
- ❑ What is the needed Value added from Investors?
- ❑ How much “skin in the game” should we have?

Reconciling The Lenses: Is There A Match?



- Is there congruence of objectives? “View of the world”? Trust?
- Complementarities of skills and capabilities among entrepreneurs and Investors? Chemistry? Respect?
- Are the MMM criteria of both the investors & entrepreneurs met?



Takeaways...

- MMM framework
 - Criteria for evaluating the market
 - Size, Acceptance, Competition, Timing
 - Criteria for evaluating the financial viability of the venture
 - Criteria for evaluating the management
 - Entrepreneurs, Team, Company,
- The “Jockey or Horse” debate
 - Weighing the evaluation of market and management

Next Class: Rent a runway Case



Jennifer Hyman



Jennifer Fleiss